Woojun Jung

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RESEARCH INTERESTS

Foundations in Agent Memory: understanding and shaping the memory representations to support reliable knowledge revision and continual adaptation.

Personalized Conversational Agents: learning from feedback to model users' stable and evolving preferences across contexts, and using uncertainty to guide personalization and active learning.

Reasoning for Social Intelligence: inferring others' beliefs, intentions, and social context by integrating verbal and nonverbal cues across interactions, to guide communication and cooperation.

EDUCATION

Korea University *Mar 2025 – Feb 2027 (Expected)*
M.S. in Computer Science and Engineering
Advisor: Prof. Susik Yoon

Korea University *Mar 2022 – Feb 2025*
B.E. in Computer Science and Engineering

Korea University *Mar 2016 – Feb 2022*
B.F.A. in Art and Design *2 years of Military Service (2017 – 2019)*

Relevant Coursework: *CS/AI* - Algorithms (A+), Computer Architecture (A), Artificial Intelligence (A), Deep Learning (A), Optimization Theory (A), Reinforcement Learning (A), Natural Language Processing (A), Brain and Machine Cognition (A+); *Design/HCI* - Interaction Design (A+), Interface Design (A+), Design and Human Factors (A+), Design Methodology (A+).

PUBLICATIONS

[3] **Segment-driven Structural Induction and Semantic Alignment for Heterogeneous Tabular Representation**
Woojun Jung and Susik Yoon
ICML 2026 — [PMLR] [Code]

[2] **Can Structural Cues Save LLMs? Evaluating Language Models in Massive Document Streams**
Yukyung Lee, Yebin Lim*, Woojun Jung*, Wonjun Choi, Susik Yoon
KDD 2026 — [ACM DL] [Code] (*: equal contribution)

[1] **HAETAE: In-Domain Table Pretraining with Header Anchoring**
Woojun Jung and Susik Yoon
SIGIR 2025 (Short) — [ACM DL] [Code]

RESEARCH EXPERIENCE

DAIS Lab, Korea University *Aug 2024 – Present*
Research Intern (Aug 2024 – Feb 2025); Research Assistant (Mar 2025 – Present)

Calibrated User Beliefs for Conversational Agents *May 2026 – Present*

- Defining long-horizon tasks for response personalization and agent action selection.
- Designing evaluation protocols to assess calibration of predicted probabilities that remembered preferences apply in new contexts, using subsequent user judgments.
- Exploring how to estimate uncertainty from learned representations and whether these estimates can improve response personalization and clarification decisions.

Structured Understanding of LLMs in Evolving Information Streams **KDD 2026**

- Developed the Temporal QA task, one of three benchmark tasks alongside Topic Clustering and Summarization.
- Designed and implemented its evaluation protocol, benchmark dataset, and human-annotation guidelines.

Structured Understanding of Tabular Data **ICML 2026; SIGIR 2025**

- Led two first-author projects on domain-specific tabular representation learning.
- Proposed header anchoring to regularize contextualized header embeddings toward context-independent anchors, stabilizing semantics across row contexts and schema variants.
- Developed a pretraining method that jointly learns schema and column contexts with entropy-routed objectives to preserve domain-coherent and instance-specific attribute semantics.

APPLIED RESEARCH COLLABORATION

Foundation Models for Enterprise Relational Data, KT (Korea Telecom) *2025 – 2026*

- Collaborated with KT on foundation-model methods for enterprise relational databases.
- Analyzed challenges in adapting tabular ML to organizational data with heterogeneous schemas.
- Prototyped a pretraining pipeline for schema-aware attribute representations for enterprise tables.
- Developed a schema-linking method that maps natural-language questions to relevant tables and columns.

SELECTED PROJECTS

Dr.Snap: AI-guided Skin Assessment — [Link] *2024*

- Designed and built a conversational skin-assessment prototype that elicits symptom details through follow-up questions informed by retrieved condition information and prior responses.
- Mapped the care journey from initial skin concerns to clinical consultation, positioning AI as pre-visit support.

Silverlining: User Research for Adaptive Lighting — [Link] *2020*

- Conducted interviews and observations of home lighting use, identifying manual adjustments during activity transitions and informing the design of a presence-responsive pendant-light concept.

Bianca: Full-Scale Interactive Refrigerator Prototype — [Link] *2020*

- Developed Arduino code and electronics and built a full-scale refrigerator prototype that maps handle-press locations to corresponding illuminated regions on the door.

TEACHING EXPERIENCE

Teaching Assistant, Data Science, Korea University *Spring 2025*

- Prepared and delivered tutorials on LaTeX and visualization with matplotlib for undergraduate students.

Teaching Assistant, Artificial Intelligence, Korea University *Fall 2025*

- Prepared and delivered a tutorial on multimodal representation learning with CLIP for undergraduate students.

AWARDS & SCHOLARSHIPS

Master's Fellowship (KRW 12,000,000), National Research Foundation of Korea (NRF) *2025–2026*
Second Place, Korea University AI Tech Day Poster Presentation *2025*
Third Place, Korea University Capstone Design Competition *2024*
Third Place, Samsung Design Competition *2019*

SKILLS

Programming: Python, C, SQL, Git, GitHub, LaTeX, HTML, CSS, JavaScript
ML & Systems: PyTorch, CUDA, Transformers, vLLM, NumPy, pandas, scikit-learn, matplotlib
Design & Hardware Prototyping: Figma, Illustrator, Photoshop, Rhinoceros, KeyShot, 3D printing, Arduino
Languages: Korean (Native), English (Fluent—TOEFL iBT 5.5/6.0)